

7-Zip

7-Zip is an open-source archiving software with really good and frequently updated algorithms that perform compression and decompression. It's the frequent updates that make 7-Zip a good benchmark. It also scales well with higher core counts so there's an added advantage with 7-Zip over other archiving software. The Intel Core i9-14900K clearly wins here over the 7950X3D. Even the Intel Core i5-14600K bests the 13600K and nearly equals the Ryzen 9 7900X.

Mozilla Kraken

Mozilla Kraken is a browser-based JavaScript benchmark that executes several real-world scenarios that everyone comes across while normally browsing the Internet. It includes a host of different use cases such as searching, audio processing, image filtering, cryptography, and JSON parsing.

Intel usually excels at Mozilla Kraken and even Octane Javascript benchmark. Both, the Intel Core i9-14900K and the Core i5-14600K beat the 7950X3D in this benchmark.

Gaming

Gaming seems to still be in AMD's kitty at the moment. While Intel held the crown for the longest time ever, the advent of the X3D processors really turned things around for AMD. While the Intel Core i9-14900K has gained about 2 per cent performance gains here, the X3D processors continue to maintain their lead. Intel has the new Application Performance Optimisation which only affects games that Intel has built profiles for so it will be a while before we see a lot more profiles and a lot more impact of the new technology. This could have been the game changer for Intel but it appears that the feature is not yet ready at launch time. We will revisit the gaming benchmarks when APO has more games under its bag.

POWER CONSUMPTION

The Intel Core i9-14900K ends up consuming about 238 Watts on average during a 10-minute intense workload

with quite a lot of instances of the processor throttling. And the peak power consumption was at 317 Watts towards the beginning of the graph that you see below. The good thing here, as has been in the past with flagships, is that the processor continues to perform consistently well across the entire duration of the test. The previous gen Core i9-13900K would consume about 245 Watts on average with spikes hitting the 300s. Both processors are based on the same Intel 7 process with the same number of Cores and Threads. Even the power limits are the same as per the spec sheet. So the two being neck and neck seems understandable.

TEMPERATURES

Straight off the bat, the Core i9-14900K jumps to hit its max temperatures. We tried using a massive air cooler as well as an AIO to see if we can avoid hitting TjMax for a lesser duration but none of what we did could help. Some of the more exotic coolers should be able to get the temperatures lower. We're not saying that it runs hot all the time, in fact, the i9-14900K would hover around 37-39 degrees on average during normal use.

INTEL CORE i9-14900K P-CORE AND E-CORE FREQUENCY MIX

To understand the clock frequencies that the P-Cores on the Intel Core i9-14900K mostly ran at during peak workloads, we also captured the clock frequencies and created a distribution chart. We can see that the P-Cores spent most of the time running at 4.9 GHz followed by 5 GHz and 4.8 GHz. For the Core i9-14900K, 6 GHz is the max Turbo clock and it spent about 2-3 per cent of the overall benchmark

On the other hand, the E-Cores spend most of their time running at 4.1 GHz. The Max Turbo frequency for the E-Cores on the 14900K is 4.4 GHz, so this is not bad at all. It's spending a lot of time giving all that it can. Also, E-Cores aren't as affected by Thermal throttling as the P-Cores are, so this is also an expected behaviour.

VERDICT

The new Intel 14th Gen processors are an improvement to the 13th Gen processors in some aspects but the improvements are not that significant. It does not have more cache memory, nor is it manufactured on a better process node and the architecture isn't new. So improvements are there but not significant enough to warrant an upgrade from 13th Gen to 14th Gen. However, if you're coming from 12th Gen or older, then the Intel 14th Gen processors are a great upgrade. Also, you'll see a lot more improvements in performance and Intel Application Performance Optimization gets more game profiles, so that's something to look forward to.

—Mithun Mohandas

SPECIFICATIONS

Core i9-14900K / KF: MSRP: \$589 | Cores / Threads (P+E): 24 / 32 (8+16) | P-Core Base / Boost Clock (GHz): 3.2 / 6.0 | E-Core Base / Boost Clock (GHz): 2.4 / 4.4 | Cache (L2+L3): 68MB (32+36) | TDP: 125W / 253W | RAM Type & MT/s: DDR4-3200 / DDR5-5600

Core i9-13900K / KF: MSRP: \$589 | Cores / Threads (P+E): 24 / 32 (8+16) | P-Core Base / Boost Clock (GHz): 3.0 / 5.8 | E-Core Base / Boost Clock (GHz): 2.2 / 4.3 | Cache (L2+L3): 68MB (32+36) | TDP: 125W / 253W | RAM Type & MT/s: DDR4-3200 / DDR5-5600

Core i7-14700K / KF: MSRP: \$409 | Cores / Threads (P+E): 20 / 28 (8+12) | P-Core Base / Boost Clock (GHz): 3.4 / 5.6 | E-Core Base / Boost Clock (GHz): 2.5 / 4.3 | Cache (L2+L3): 61MB (24+30) | TDP: 125W / 253W | RAM Type & MT/s: DDR4-3200 / DDR5-5600

Core i7-13700K / KF: MSRP: \$409 | Cores / Threads (P+E): 16 / 24 (8+8) | P-Core Base / Boost Clock (GHz): 3.4 / 5.4 | E-Core Base / Boost Clock (GHz): 2.5 / 4.2 | Cache (L2+L3): 54MB (24+30) | TDP: 125W / 181W | RAM Type & MT/s: DDR4-3200 / DDR5-5600

Core i5-14600K / KF: MSRP: \$319 | Cores / Threads (P+E): 14 / 20 (6+8) | P-Core Base / Boost Clock (GHz): 3.5 / 5.3 | E-Core Base / Boost Clock (GHz): 2.6 / 4.0 | Cache (L2+L3): 44MB (20+24) | TDP: 125W / 181W | RAM Type & MT/s: DDR4-3200 / DDR5-5600

Core i5-13600K / KF: MSRP: \$319 | Cores / Threads (P+E): 14 / 20 (6+8) | P-Core Base / Boost Clock (GHz): 3.5 / 5.3 | E-Core Base / Boost Clock (GHz): 2.6 / 3.9 | Cache (L2+L3): 44MB (20+24) | TDP: 125W / 181W | RAM Type & MT/s: DDR4-3200 / DDR5-5600

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